Geometry & Measurement

Section 1: Angles & Triangles

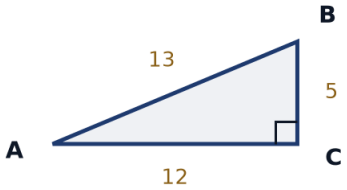
1 In a triangle, the angles are in the ratio $2:3:5$. What is the measure of the largest angle?

a. 90°

b. 100°

c. 80°

d. 75°



2 In a right triangle, one leg is $12\ \text{cm}$ and the hypotenuse is $13\ \text{cm}$. What is the area of the triangle?

a. $30\ \text{cm}^2$

b. $60\ \text{cm}^2$

c. $39\ \text{cm}^2$

d. $25\ \text{cm}^2$

3 Two parallel lines are intersected by a transversal. If an interior angle measures $(3x - 15)^\circ$ and its consecutive interior angle measures $(2x + 20)^\circ$, what is x?

a. 35

b. 30

c. 40

d. 25

Section 2: Circles, Circumference & Area

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4 A circle has a circumference of 16π cm. What is its area?

a. 64π cm^2

b. 32π cm^2

c. 16π cm^2

d. 128π cm^2

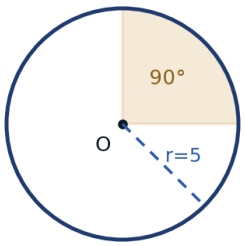
5 A square of side length 10 cm has a circle inscribed within it. What is the area of the shaded region between the square and the circle?

a. $100 - 25\pi$

b. $100 - 10\pi$

c. $50 - 25\pi$

d. $100 - 50\pi$



6 If the radius of a circle is doubled, by what factor does its area increase?

a. 4

b. 2

c. 8

d. 16

Section 3: Quadrilaterals & Polygons

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7 A rectangle has a perimeter of $40\ \text{cm}$ and its length is 3 times its width. What is the area of the rectangle?

a. $75\ \text{cm}^2$

b. $80\ \text{cm}^2$

c. $100\ \text{cm}^2$

d. $60\ \text{cm}^2$

8 What is the sum of the interior angles of a regular hexagon?

a. 720°

b. 540°

c. 360°

d. 900°

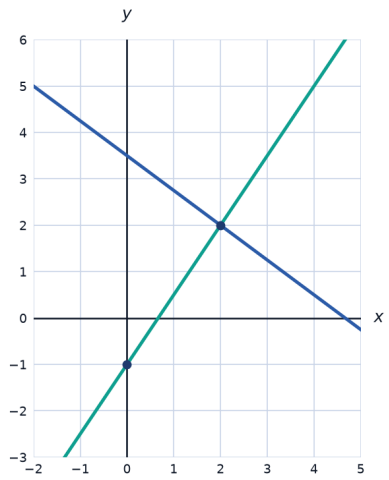
9 A rhombus has diagonals of lengths $12\ \text{cm}$ and $16\ \text{cm}$. What is the length of one side of the rhombus?

a. 10 cm

b. 8 cm

c. 14 cm

d. 20 cm



Section 4: Solid Geometry & Volume

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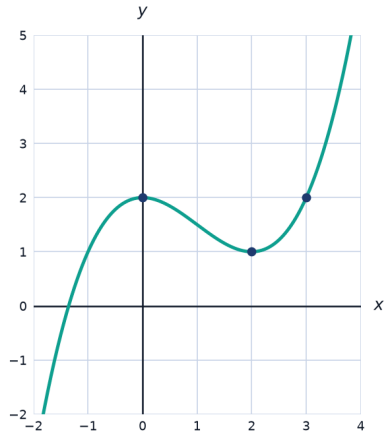
10 A rectangular tank has dimensions $4\text{ m} \times 5\text{ m} \times 6\text{ m}$. If it is filled with water at a rate of $10\text{ m}^3/\text{min}$, how many minutes does it take to fill completely?

a. 12 minutes

b. 10 minutes

c. 15 minutes

d. 20 minutes



11 A cylinder has base radius 3 cm and height 10 cm. What is its volume in cubic centimeters?

a. 90π

b. 60π

c. 30π

d. 180π

12 If the volume of a cube is 64 cm^3, what is its total surface area?

a. 96 cm^2

b. 64 cm^2

c. 128 cm^2

d. 48 cm^2

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